TrES-5b

R-0985

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_200916` · created 2026-07-29T22:06:00+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459108.8511 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.04
Transit depth (Rp/Rs)^2	3.4 +/- 1.46 [%]
Orbital Inclination (inc)	88.83 +/- 2.27
Airmass coefficient 1 (a1)	1.06 +/- 0.033
Airmass coefficient 2 (a2)	-0.022 +/- 0.026
Scatter in the residuals of the lightcurve fit is	3.17 %
Best Comparison Star	#3 - [287, 341]
Optimal Aperture	2.83
Optimal Annulus	9.84
Transit Duration (day)	0.0655 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.04 vs 0.142 (deviation 1.03σ)
Duration fit vs published	0.0655 d vs 0.0758 d (deviation 2.9σ)
Mid-time O–C	2.9 min
Depth SNR	4.5
Residual scatter	3.17 %

Light curve

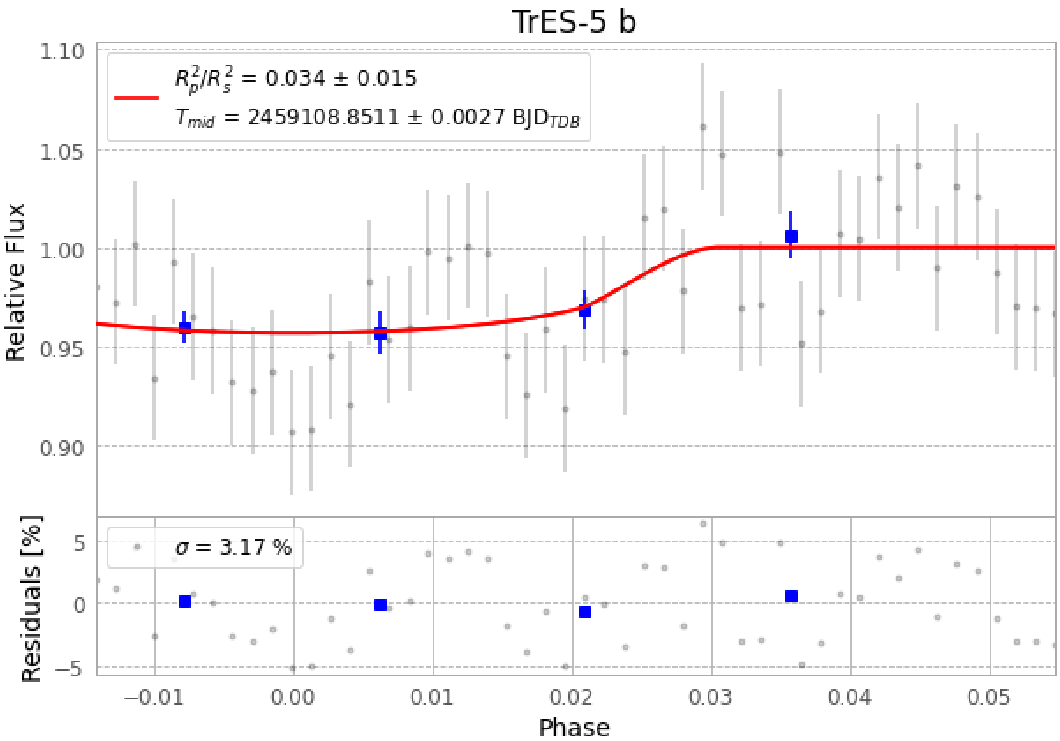

Figure 1 — FinalLightCurve_TrES-5 b_2020-09-16.png (SHA-256 a6832a344ac27dbd…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [382, 307], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d7a7c7419566f89a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

50 FITS frames (33 MB), `TRES-5200916075113.FITS` → `TRES-5200916101815.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-200916.log (515a11a8fa3e…), FinalLightCurve_TrES-5 b_2020-09-16.png (a6832a344ac2…), FinalLightCurve_TrES-5 b_2020-09-16.csv (924662c71527…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)